3D printing Instructions

Checkout our CURA 3mf examples for more detailed parameters:

https://drive.google.com/drive/folders/15vFeCd-fEt-NOYkRXebhpDGm5D2zX3MM?usp=sharing

3D Printed Parts List

Part Name	Quantity	Material	Category	Notes
Bottom Plate	1	PLA+	Main Structure	Main lower structural plate
Cube	1	PLA+	Main Structure	Central structural block
Finger Holder Left	1	PLA+	Finger Holder	Left finger support
Finger Holder Right	1	PLA+	Finger Holder	Right finger support
Gear Left	1	PLA+	Motion Transmission	Left actuation gear
Gear Right	1	PLA+	Motion Transmission	Right actuation gear
Grip	1	PLA+	Main Structure	Main hand-contact body
Handle	1	PLA+	Main Structure	Handle body
Linkage Left	1	PLA+	Motion Transmission	Left linkage element
Linkage Right	1	PLA+	Motion Transmission	Right linkage element
Top Cover 2026	1	PLA+	Main Structure	Upper cover plate
Hard Finger 1	1	PLA+	Interchangeable Finger	Left/right hard-finger pair component
Hard Finger 2	1	PLA+	Interchangeable Finger	Left/right hard-finger pair component
Soft Gripper Finger 1	1	TPU 95A	Interchangeable Finger	Left/right soft-finger pair component
Soft Gripper Finger 2	1	TPU 95A	Interchangeable Finger	Left/right soft-finger pair component

Note

Only one interchangeable finger pair is used in the final assembly at a time:

- either the soft-finger pair,
- or the hard-finger pair.

3D Printing Guidelines

Common Print Parameters for PLA+ Parts

The following settings were used as the common baseline for all PLA+ printed components unless otherwise specified:

- Material: eSun PLA+
- Nozzle diameter: 0.6 mm
- Layer height: 0.3 mm
- Wall thickness: 1.2 mm
- Top-bottom thickness: 1.2 mm
- Infill: 20% Gyroid
- Printing temperature: 190°C
- Cooling fan: 100%

Finger Print Parameters for TPU Parts

The following settings were used for the flexible finger components:

- Material: Polymaker TPU 95A
- Nozzle diameter: 0.6 mm
- Layer height: 0.3 mm
- Infill: 100% Lines
- Printing temperature: 255°C
- Cooling fan: 0%

Part-Specific Printing Table

Print Group	**Part**	**Material**	**Infill %**	**Support Required**	**Wall Thickness (mm)**	**Top-Bottom Thickness (mm)**	**Notes**
Print 1: Gripper Covers	Bottom Plate	PLA+	20	No	1.2	1.2	Use common PLA+ print settings
	Top Cover 2026	PLA+	20	No	1.2	1.2	Use common PLA+ print settings
Print 2: Gripper Internals	Grip	PLA+	100	No	1.8	1.2	Increased infill for higher structural strength
	Handle	PLA+	20	No	1.2	1.2	Use common PLA+ print settings
	Cube	PLA+	20	No	1.2	1.2	Use common PLA+ print settings
	Finger Holder Left	PLA+	20	Yes	1.2	1.2	Print with supports
	Finger Holder Right	PLA+	20	Yes	1.2	1.2	Print with supports
	Gear Left	PLA+	20	No	1.2	1.2	Use common PLA+ print settings
	Gear Right	PLA+	20	No	1.2	1.2	Use common PLA+ print settings

	Linkage Left	PLA+	20	No	2.4	1.8	Increased wall and top-bottom thickness
	Linkage Right	PLA+	20	No	2.4	1.8	Increased wall and top-bottom thickness
Print 3: Gripper Fingers	Hard Finger 1	PLA+	100	No	1.2	1.2	Rigid finger configuration
	Hard Finger 2	PLA+	100	No	1.2	1.2	Rigid finger configuration
	Soft Gripper Finger 1	TPU 95A	100	No	—	—	Print with TPU finger settings
	Soft Gripper Finger 2	TPU 95A	100	No	—	—	Print with TPU finger settings

3D Printing Guidelines

The following guidelines summarize the recommended workflow for fabricating the modified UMI components.

1. Select a reliable FDM 3D printer capable of printing PLA+ and TPU materials.
2. Install a 0.6 mm nozzle to reduce print time and improve structural robustness.
3. Level the print bed carefully to ensure proper first-layer adhesion.
4. Import the provided .3mf or .stl files into the slicer software (e.g., Cura).
5. Apply part-specific print settings according to the component category (structure, linkage, fingers).
6. Use standard PLA+ settings for all structural components.
7. Use dedicated TPU settings (low cooling, high infill) for soft-finger components.
8. Adjust tolerances in CAD if necessary to ensure proper fit between moving parts.

9. Customize the hand grip geometry in CAD if user-specific ergonomics are required.
10. Prepare the printer by cleaning the build plate and verifying material loading.
11. Start the print and monitor the first layers to confirm correct adhesion.